

RESEARCH ARTICLE

Remote Sensing Surveillance Using Multilevel Feature Fusion and Deep Neural Network

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ABSTRACT Human action recognition from aerial imagery poses significant challenges due to the dynamic nature of the scenes and the complexity of human movements. In this paper, we present an enhanced system that combines YOLO for human detection with a complete multilevel feature fusion approach to improve recognition of human actions in drone-captured photos. Our system presents a reliable drone-based human action system through the integration of state-of-the-art methods for multilevel feature extraction and object detection. Initially, frames are extracted individually from drone footage sequences. Preprocessing techniques, which include Gaussian blur, grayscale conversion, and background removal, are used for every frame to improve image quality and feature reliability. For object detection, we effectively locate and recognize human subjects in these aerial frames using YOLO approach. Afterward, the framework extracts 14 body landmarks that represent the shape of the human body by keypoint extraction. Four significant features are employed to capture the complexity of human movement effectively: the incorporation of 3D point cloud data adds depth to the image and makes it feasible to construct a more detailed three-dimensional representation; measuring the angles between keypoints provides significant details on joint orientations which are essential for posture analysis; and geodesic distance measure the shortest paths along the surface of the body to provide useful insight into the spatial relationships between keypoints. The extracted features are optimized by using quadratic discriminant analysis. In the end, a deep neural network was trained to perform the action classification. Three benchmark datasets, the UAV Gesture, UAV Human, and UCF-ARG datasets, were used for our experiments and system testing. Our model achieved corresponding action recognition accuracy values of 90.15%, 72.37%, and 76.50% on each of these datasets.

INDEX TERMS Human action recognition, object detectors, aerial imaging, body pose, multilevel feature fusion, image analysis, deep learning.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are being employed increasingly in the modern era for a variety of functions

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including sports, entertainment, security, crowd surveillance, search and rescue, and monitoring of airborne human activity. UAVs are more flexible in this regard than security cameras because fixed cameras have preset angles and dimensions that cannot be altered [1]. Many researchers have recently accepted that HAR is a crucial component of numerous

applications including search and rescue to monitoring [2]. The benefit of UAV-based HAR systems is that they may be put in places with insufficient connectivity and cover a wide region. Given society's growing dependency on these kinds of technology, human actions must be able to be accurately identified and categorized promptly [3], [4]. The majority of traditional human action detection paradigms take advantage of static camera footage, even though its action recognition abilities are limited to certain viewpoint angles. UAVs cover a large region that ground-based systems are unable to cover [5], [6]. These devices are quite adaptable since they can capture still images and videos from practically any angle or height [7]. This capability could be beneficial for a variety of purposes, including disaster relief, military applications, and surveillance for large-scale and magnitude monitoring.

Difficulties are likely to occur even though the task is done with drone footage. This is usually the case due to UAV's unpredictable nature, making it hard to extract reliable and accurate information from the photographs they capture. It is also challenging to distinguish movements and identify the human in a lot of drawn footage because of the dynamic, varying surroundings. Furthermore, the motion of the UAV combined with the frequent change in size, angle, and display of different human figure images makes it challenging to maintain a good separation and consequently the monitoring of frames about degrees of recognition. A more sophisticated background [8], also seems to be present, which makes it harder to tell one individual apart in unpredictable or complex situations. Because of these characteristics, a more complex technique is required to guarantee proper identification of human activities, especially considering the challenging conditions in which drone footage is found. This research provided a new framework for human activity recognition specifically designed for drone images to address these problems.

We employ multilevel feature fusion and apply image improvement techniques to improve the resilience and precision of the action recognition in complex aerial circumstances. Finally, we use a neural network approach for classification. Therefore, frame extraction is the initial stage in which the algorithm splits the videos into frames to process the continuous video data individually. Following frame extraction, the following stage seeks to improve image quality and organize the data to improve this analysis. Gaussian blur, grayscale conversion, and background removal are required preprocessing methods, among others. Gaussian blur is used to perform the other procedures, which include noise reduction and smooth images without losing crucial details. Next, before the color information gets processed, the entire image is converted to grayscale, removing a lot of unwanted data and focusing on the structures and background to remove the noise. We use the YOLO model, which can reliably identify humans in the frame even in difficult situations. YOLO (You Only Look Once) is a popular and robust method for object detection that is highly reliable and efficient.

YOLO creates the bounding boxes and class probabilities for the entire image in a single shot, as compared to earlier region-based detection methods. This combined method significantly reduces computational overhead while maintaining detection accuracy. YOLO's architecture facilitates effective feature extraction and localization, it can detect objects with exceptional precision at a variety of scales and orientations.

Moreover, YOLO can handle several objects in a single image. It incorporates contextual information from every image, reducing false positives and maintaining its robustness in difficult situations like cluttered or crowded scenes. Autonomous vehicles, medical imaging, and security systems are just a few of the applications that can benefit greatly from YOLO's adaptability to various datasets.

When used in the context of UAV-based human action recognition, YOLO has been demonstrated to have some benefits that might help resolve issues from an aerial perspective. The unpredictable nature of drone footage taken from varied angles, at varying heights, and in variable lighting conditions may also be difficult for humans to handle. Given its capacity to recognize individuals at various scales and angles, it is possible to show that YOLO ensures precise human identification in these situations. Drone images contain complex backgrounds such as natural landscapes or urban environments, where distinguishing human figures is challenging. That is why using a global image context YOLO minimizes distractions and can isolate human subjects effectively. YOLO allows for detecting human subjects in the image. Moreover, it provides highly reliable performance even in low conditions and is suitable for further analysis such as the extraction of keypoints and action classification. The integration of YOLO makes the proposed system more robust thereby allowing accurate and efficient human detection in areas such as security, search and rescue, and disaster management.

Apart from its application in UAV-based human action recognition, YOLO is being used to address real-world issues across various fields, like sports and the medical domain. In the domain of sports, YOLO is used to improve both officiation and performance analysis. For instance [9], used YOLOv5 to develop an automated badminton match fault detection system for service height. The system identifies shuttlecocks, rackets, and players, restricting the shuttlecocks to the legal limits while offering a far more equitable system in general. In addition to reducing human error in decision-making, these applications offer information about player performance. In various sports contexts, YOLO has been applied similarly to sports engineering. For instance, analyzing player movements and identifying activities allows the tools to provide coaching and strategy development. YOLO has demonstrated efficacy in the medical field in improving the precision and effectiveness of clinical procedures.

YOLO has also been used to identify sports referee gestures. A lightweight, high-performing YOLO-based model called FRGR-YOLOv8s is proposed in a [10] study for

the recognition of football referee gestures. This approach tackles the typical issues of low lighting and complicated backgrounds or occlusions that are frequently present in sporting settings. In order to increase game management's effectiveness and dependability, the FRGR-YOLOv8s model precisely recognizes and categorizes referee gestures. It also exhibits YOLO's resilience under demanding and dynamic circumstances.

One specific example is the use of YOLO for 3D posture monitoring of cardiac catheters during bi-plane fluoroscopy [11]. This technique significantly improves the precision of minimally invasive procedures by precisely detecting important landmarks like catheter tips and entry locations. Furthermore, YOLO has been used for medical imaging activities such as posture analysis, tumor segmentation, and lesion recognition, which gives it the potential to be an automated tool for medical professionals for intricate diagnosis and treatment procedures.

After that, 14 body landmarks are identified using point extraction, including the nose, neck, right shoulder, left shoulder, right elbow, left elbow, right wrist, left wrist, left hip, right hip, right knee, left knee, right ankle, and left ankle. This provides the skeletal model that is needed for subsequent analysis. We employ multiple-level feature fusion for identification and improvements of the essential features such as joint angles, geodesic distance, 3D point cloud, and MSER to properly represent human actions. A more accurate and refined feature set is obtained after these features have been improved by applying Quadratic discriminant analysis. Subsequently, a deep neural network classifier is utilized for precise recognition of human motions with great resilience of signals recorded even in the presence of other movements. Our method attains greater action recognition accuracy compared with past algorithms in action recognition while integrating multilevel feature fusion with a neural network approach and accurate human detection suitable for applications that need aerial imagery. The important contributions to our study are given below:

- We applied YOLO for human detection efficiently recognizing human subject, enabling correct detection of humans.
- The system uses multilevel feature fusion techniques, including the angle between joints, geodesic distance, 3d point cloud, and MSER to give an in-depth study of human motion.
- Our technique increases action categorization by employing DNN.

The rest of the paper is organized as follows: Section II provides a literature review of existing methods. Section III then discusses the proposed system. Section IV presents the experimental design and findings. Section V examines the suggested system's effectiveness over the datasets and the insights obtained from the experiments. Section VI concludes with conclusions and suggestions for additional research.

II. LITERATURE REVIEW

In computer vision, researchers have utilized fixed cameras to capture, detect, and interpret human behaviors using various techniques such as skeleton-based, optical flow-based, Hidden Markov Model, and spatiotemporal feature extraction. While these methods provide advantages like simplicity and low computational cost, they suffer from significant limitations, including sensitivity to viewpoint, lack of temporal context, and constraints to specific scenarios.

With advancements in drone technology, researchers have shifted their focus from stationary cameras to aerial views for human behavior detection. UAV-based Human Action Recognition (HAR) refers to the detection and classification of human behaviors from above using Unmanned Aerial Vehicles (UAVs) or aerial robots. UAV-based HAR has evolved significantly, starting with basic motion and action recognition and progressing through deep learning techniques (such as CNNs and RNNs) to more advanced methods like multi-modal sensor fusion and real-time HAR. These advancements have enabled UAV-based HAR to be applied across various domains, such as search and rescue operations [12], pedestrian monitoring [13], and agriculture [14]. However, UAV-based HAR still faces critical challenges, such as image quality, limited field of view, occlusion, and environmental factors.

A. OBJECT DETECTION ALGORITHMS IN UAV-BASED HAR

Human action recognition in aerial imagery from UAVs presents unique challenges compared to traditional object detection tasks, including varying altitudes, perspectives, and the small size of objects within images. Recent deep learning advancements, especially in object detection algorithms, have introduced robust solutions. This section reviews recent developments in object detection algorithms specifically for UAV contexts, focusing on methods like YOLO and Faster R-CNN. A study in [15] examined the effectiveness of detectors like Faster R-CNN, YOLOv4, RetinaNet, and Cascade R-CNN using the VisDrone benchmark dataset and a custom search and rescue dataset (SARD), finding that YOLOv4 outperformed other detectors, making it highly suitable for aerial platforms. The growing availability of aerial visual data offers tremendous potential for object detection, significantly enhancing UAV capabilities in applications such as automated flying, disaster management, weather forecasting, and military operations.

Recent studies on human action recognition and anomaly detection in drone images highlight how tailored deep-learning models can improve UAV system intelligence. For instance, Maheriya et al. [16] proposed an adaptive fusion mechanism, which achieved a 7.1% improvement in small object detection for UAV scenarios compared to YOLOv5. Similarly, Xian et al. [17] found that YOLOv4 outperformed other detectors, underscoring its suitability for UAV applications. Literature indicates that YOLO is ideal for

real-time action recognition in aerial imagery due to its accuracy and efficiency in predicting multiple bounding boxes and class probabilities simultaneously [16]. Its grid-based convolutional architecture captures spatial hierarchies and contextual information, which is especially useful for detecting smaller human figures in aerial images and handling scale and perspective variations. YOLO's robustness to lighting and occlusion further supports its application in surveillance, search and rescue, and urban planning.

B. FEATURE EXTRACTION AND MULTILEVEL FEATURE FUSION TECHNIQUES

Body landmark extraction from aerial images is crucial for understanding human posture and movement. Studies have applied both traditional computer vision and deep learning methods to accurately identify human body key points, a capability vital for applications like surveillance and sports analysis. This technology enables reconstruction of human posture and complex movement tracking, thereby enhancing situational awareness in various contexts [18].

Multilevel feature fusion enhances HAR by integrating information from sources such as 3D point cloud data, joint angles, and geodesic distances. This approach yields a more comprehensive representation of human activities, enabling models to better differentiate similar actions and thereby improving recognition accuracy [63]. Techniques like Quadratic Discriminant Analysis (QDA) refine features, increasing classification robustness by defining quadratic boundaries around class distributions. Similarly, Principal Component Analysis (PCA) optimizes the feature space, further enhancing classification performance in HAR tasks.

C. DEEP LEARNING APPROACHES FOR HAR

Deep Neural Networks (DNNs) have become indispensable in HAR due to their capacity to learn complex patterns from large datasets. A recent study by Zhang et al. demonstrates the effectiveness of DNNs for HAR, using a multi-layer architecture to extract spatial and temporal features from wearable sensor data, such as accelerometer and gyroscope readings [79]. Their findings indicate that the DNN model outperforms traditional machine learning techniques, achieving higher accuracy in recognizing activities like walking, running, and sitting, and showing strong generalization across different subjects and settings.

In another approach, Li et al. combine Quadratic Discriminant Analysis (QDA) with deep learning to improve feature classification [80]. Their hybrid model leverages the representational power of DNNs with the statistical rigor of QDA, enhancing discrimination between activities, particularly in datasets with overlapping activities. Their results demonstrate that this combination not only boosts classification accuracy but also reduces misclassifications, making it particularly effective for differentiating between similar activities like walking and jogging.

Addressing these gaps opens numerous avenues for advancing UAV-based HAR. Integrating multilevel feature fusion with advanced neural networks offers a promising approach for capturing more nuanced representations of human actions. By combining information from diverse modalities (such as optical flow, 3D point cloud data, and body landmarks), models can achieve a richer and more comprehensive understanding of actions. Moreover, optimizing detection and recognition frameworks for real-time processing is essential, as it would enable UAVs to perform more complex tasks with immediate responsiveness. Enhancements in robust feature extraction and fusion techniques, tailored to the challenges of aerial imagery, present opportunities to significantly improve accuracy and reliability across various applications.

III. THE PROPOSED SYSTEM

The proposed system for human action recognition using drone footage goes through several sequential steps to properly process and evaluate the data. First, the drone's recorded video footage is processed to extract frames, which are input for further processing operations. Preprocessing of the extracted frames consists of converting them to greyscale to minimize complexity and applying Gaussian blurring to eliminate noise and enhance image quality. Background noise is reduced by background removal, after which the YOLO model is employed for human detection. YOLO locates and identifies people by generating bounding boxes around observed individuals, enabling effective detection of dynamic drone footage. keypoints were extracted from the detected human figures after humans had been identified. Specifically, 14 body landmarks—including the nose, neck, right shoulder, left shoulder, left elbow, right elbow, right wrist, left wrist, left hip, right hip, right knee, left knee, left ankle, and right ankle are the focus of this keypoint extraction process. The system extracts four features: the angles formed by body joints to capture posture; the angle between adjacent keypoints within a frame; geodesic distances between body landmarks to highlight spatial relationships between keypoints within a frame; a 3D point cloud representing the human figure using the point cloud data from the RGB image and the corresponding silhouette; and maximally stable extremal regions for steady region identification. to improve the accuracy of the extracted features classification. The data is subjected to the QDA-guided optimization procedure. Lastly, the DNN receives the optimized data to execute the classification. The architecture of the proposed system is illustrated in Figure 1.

A. PRE-PROCESSING

Preprocessing is an essential step in the proposed approach because drone videos are used to identify human actions. Several preprocessing techniques that enhance the frames for efficient feature extraction and analysis come beyond the initial step of frame extraction. Frame extraction is used to split video sequences into individual frames [19]. The next step involves preprocessing the frames to remove noise and

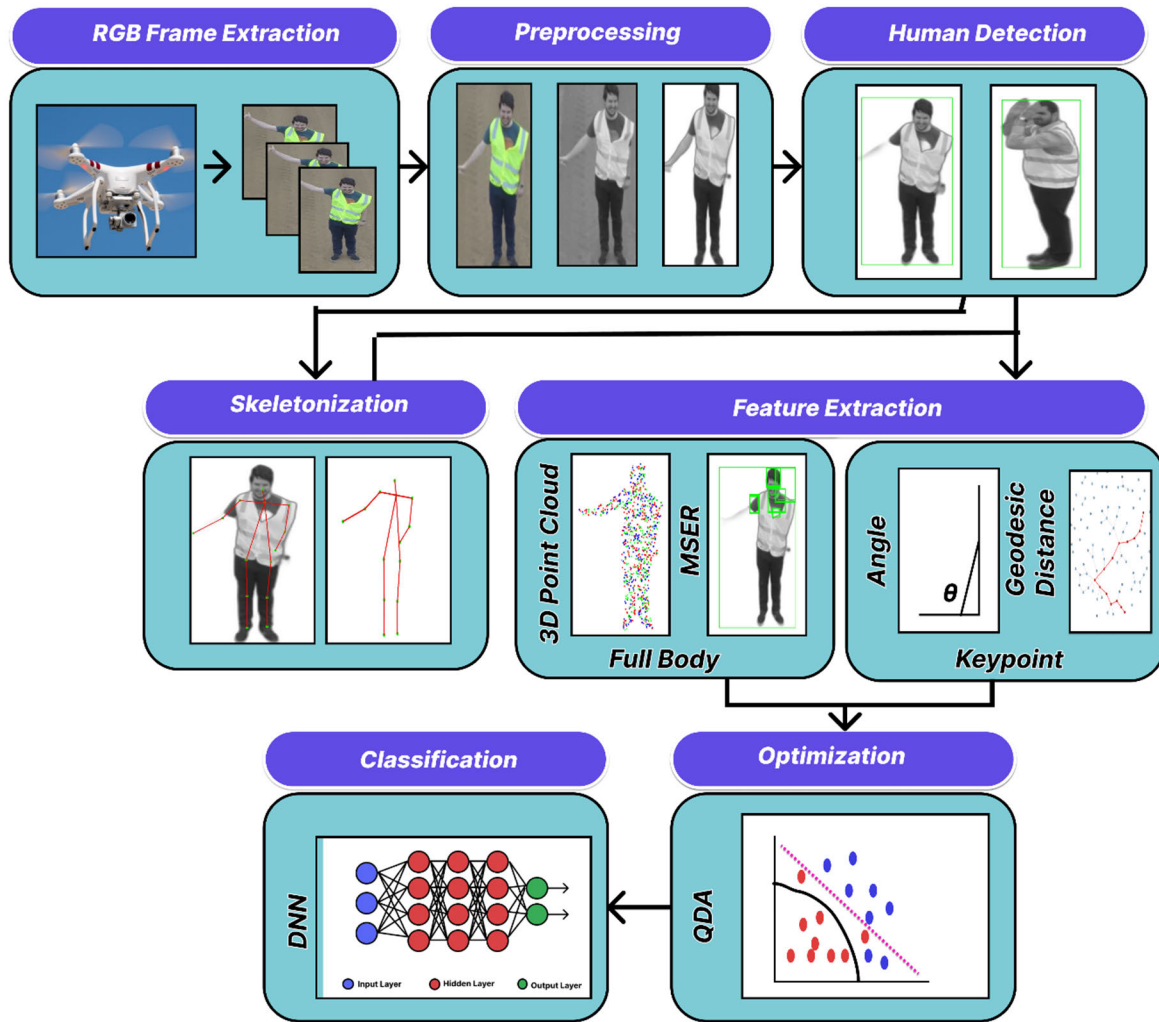


FIGURE 1. The architecture of the proposed system for human action recognition.

background components to improve the overall quality of the data [20]. The previously mentioned procedures, which serve different roles in the data preparation procedure for feature extraction, consist of Gaussian blur, grayscale conversion, and background removal [21]. A Gaussian Blur filter is used to improve frame quality and reduce noise. The image is smoothed with Gaussian blur by averaging pixel values with adjacent pixels [22]. The Gaussian kernel, $G(x, y)$, defined in Equation (1), is used for convolution to reduce noise while preserving edges:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (1)$$

The vertical and horizontal centers of the kernel are denoted by x and y , accordingly. The Gaussian function is denoted by $G(x, y)$, and since the mathematical value is exp, the standard deviation, or σ , controls the variance of the Gaussian distribution [7]. The image is simply blurred during this procedure,

with the Gaussian distribution's standard deviation parameter determining how much blurring happens. This method successfully minimizes high-frequency components like noise while maintaining critical features like edges, enhancing the clarity of the human figure [23]. Following Gaussian blur, grayscale conversion is the next preprocessing step. An RGB image must be converted to grayscale, with each pixel represented by a single intensity value. Average and weighted approaches are common techniques for converting images to grayscale. The grayscale value can be calculated using Equation (2):

$$\text{Grayscale} = 0.299R + 0.587G + 0.114B \quad (2)$$

Through this conversion, the dimensionality of the data is reduced from three to one, and concentrating exclusively on the intensity variations inside the image improves the performance of future operations, including background subtraction and feature extraction [24]. Later phases of the system can

operate more efficiently because it can process the frames faster and more effectively by removing color information. During the preprocessing pipeline, background removal is essential for isolating the human subject from the surroundings to remove the noise [25]. The background noise in UAV-based datasets often includes elements such as trees, buildings, and vehicles, can interfere with accurate detection and feature extraction. These elements might introduce noise throughout the frames, leading to false positives during feature extraction, keypoint extraction, or human detection. By removing the background, the human subject is localized and the other components are eliminated, leaving only the area of interest. In addition to improving human identification across the frame, it offers clear, targeted areas for feature extraction, such as geodesic distance computation or other key point analysis. Also, the removal of background elements reduces the computational process, and hence the processing time of captured frames from a drone is much faster and more efficient. This improves detection accuracy, ensures accurate feature extraction, and reduces computational overhead, leading to faster processing. This background removal improves the accuracy of subsequent person detection and feature extraction phases by focusing on analyzing human actions captured in the drone footage [26]. By combining these techniques, the preprocessing pipeline ensures that the data is clear, focused, and generates frames optimal for additional feature extraction, like keypoint extraction or shape analysis. Figure 2 displays the original frame and frames after preprocessing techniques.

B. HUMAN DETECTION

A cutting-edge object recognition technique was developed to tackle the difficulties of high accuracy and efficiency in real-time object detection [20]. Unlike previous approaches of object identification, YOLO generally incorporates bounding box regression, object classification, and region suggestions [27], all of which are performed in one pass in this model. This end-to-end solution, while giving lower levels of accuracy, is helpful for applications that need to operate in real-time as it considerably improves processing and makes the detection process easier.

For every cell in the grid, a particular number of bounding boxes together with their confidences and class Predictions should be provided. It also allows YOLO to handle object recognition straightforwardly as each grid cell strives to compute the bounding box and the accompanying scores from the picture itself [28]. Consequently, YOLO is particularly helpful in identifying persons when it comes to drone imagery in our research. Since drone footage is likely to be dynamic recognition of human becomes highly crucial [29]. For the same reason that we need to monitor human from above in drone footage, YOLO real-time picture processing capacity is vital for usage in surveillance systems or search and rescue operations.

In each grid cell (i, j) , YOLO exactly calculates the boxes and class probabilities in the same format of (i, j) . The height

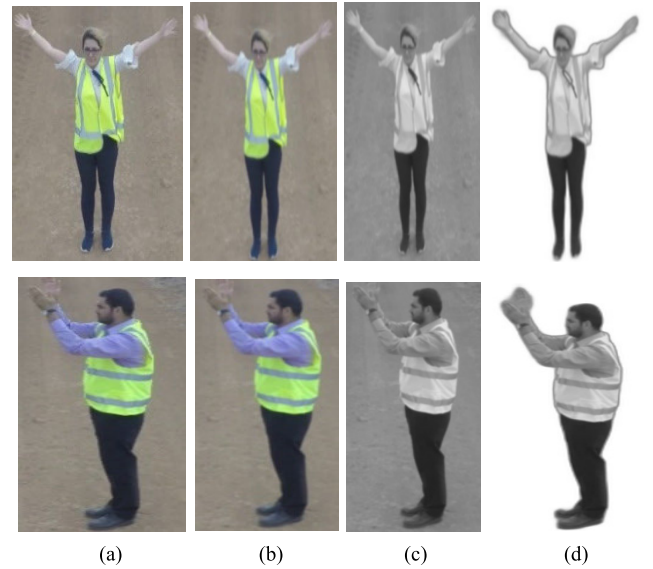


FIGURE 2. Preprocessing pipeline for UAV Gesture dataset. (a) Original images, (b) Gaussian blur, (c) Grayscale conversion, and (d) Background removal.

$h_{i,j,b}$, center coordinates $(x_{i,j,b}, y_{i,j,b})$, and width $w_{i,j,b}$ are the four parameters that each grid cell uses to build its B bounding boxes. The bounding box confidence score is calculated using Equation (3) to assess the likelihood of object presence and its overlap with the ground truth bounding box:

$$C_{i,j,b} = P_{object} \times IoU_{pred}^{truth} \quad (3)$$

The probability that the bounding box contains an object is denoted by P_{object} and the point of intersection over the union among the bounding box that was predicted and the ground truth bounding box is represented by IoU_{pred}^{truth} . The probability of an object belonging to class c_k is computed using Equation (4):

$$P_{i,j,b}(c_k) = P_{object} \times P(c_k|object) \quad (4)$$

where $P(c_k|object)$ denotes the probability of class k given the presence of an object [30]. In our work, drone footage is analyzed using YOLO's real-time human detection and classification capabilities. Even in more complicated and varied landscapes captured by drones, we can effectively recognize human forms because of YOLO's unified detection architecture [31]. Accurate class labels and bounding box coordinates are included in the output, making identifying humans easier and allowing for further research and action. The effectiveness of YOLO in detecting and classifying humans from drone images is illustrated in Figure 3.

C. KEYPOINT EXTRACTION AND SKELETONIZATION

In the keypoint extraction phase, the bounding box coordinates of every human that was detected using the YOLO model in the preceding step are the input. To facilitate the extraction of crucial points on the human body in the subsequent phase [32], these bounding boxes establish the region of

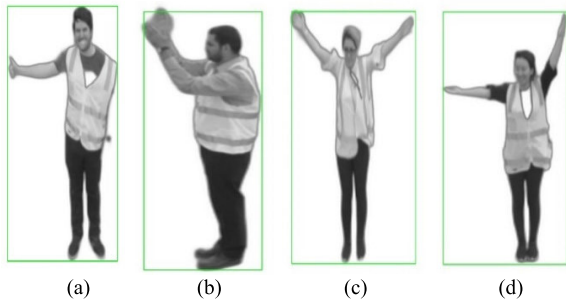


FIGURE 3. YOLO results on the UAV Gesture dataset for four different human action classes: (a) All clear, (b) Landing direction, (c) Wave off, and (d) Move to left.

interest (ROI). Because of this exact localization, the method is more accurate and efficient because it guarantees that the pose estimation is applied to the persons that have been recognized [33]. Identifying particular, anatomically significant points on the human body—also referred to as landmarks—is known as keypoint extraction [34]. The landmarks that demonstrate a structural representation of the human body are the nose, neck, left shoulder, right shoulder, left elbow, right elbow, left wrist, right wrist, right knee, left knee, left hip, right hip, left ankle, and right ankle, are illustrated in Figure 4. This approach grows on the MediaPipe pose estimation procedure and detects 14 landmarks on a human's body. From the input image, the model can correctly determine the coordinates of each of the keypoint [35]. These landmarks are verified by the MediaPipe pose model, where spatial patterns in the picture have been identified and then the model figures out how exactly these locations are located. About the human motions and posture, these landmarks convey important and supplementary information. Therefore, skeletonization connects those mentioned keypoints to develop a more skeletal foundation of the human body [36]. It assists in determining and assessing the movements of an individual in their skeletal system. Thereby 'screened' combinations of indices are used to link some of the keypoints, and generate lines corresponding to the human body's bones. By integrating those computed locations into the skeleton [37], it is possible to represent human joints and body parts by drawing lines from landmarks.

The model generates a skeleton representation of the human figure that is both computationally efficient and visually apparent by drawing lines between pairs of keypoints based on predetermined connections. For each keypoint P_i , its coordinates are represented as; $P_i = (x_i, y_i)$, where P_i is the i th keypoint, x_i is the horizontal position and y_i is the vertical position. We derive new keypoints as midpoints between two existing points to better capture human body structure. The formula for the midpoint is shown in the Equation (5):

$$P_m = \left(\frac{x_i + x_j}{2}, \frac{y_i + y_j}{2} \right) \quad (5)$$

Between two keypoints, the variable P_m in Eq (5) represents the midpoint derived from the mathematical modeling

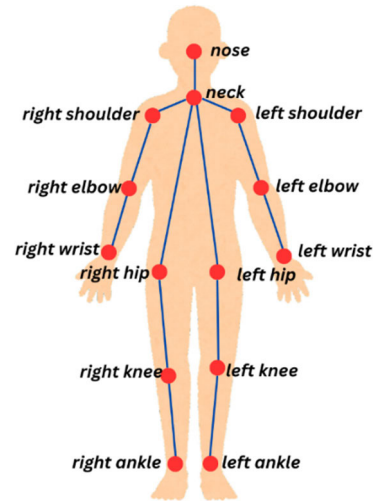


FIGURE 4. Labeled skeleton keypoints used in the proposed human recognition system.

of the human skeleton. For example, it may be interpreted as a standing midpoint between the left and right shoulder, elbow, wrist, knee, or ankle. The addition of important spatial locations that define more precise relationships between two body components makes these midpoints beneficial for enhancing the skeletal representation as well. Because keypoints have coordinate vectors, the model introduces midpoints between keypoints to build structural information of the human body.

When defining the position of the limbs, for instance, the mid-point between the elbows and knees works better since it makes it easier to track the body's motions and postures by precisely defining the angles and locations of the arms and legs. This representation is crucial for comprehending people's actions and interactions since action analysis relies on the relative positions and orientations of the keypoints.

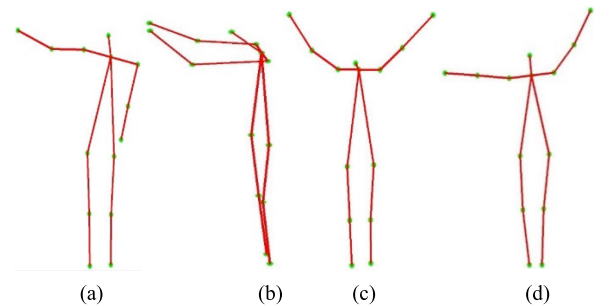


FIGURE 5. Keypoint extraction with four different human action classes of the UAV Gesture dataset: (a) All clear, (b) Landing direction, (c) Wave off, and (d) Move to left.

The keypoint extraction and skeletal modeling procedure for four different human movement classes from the UAV Gesture dataset is shown in Figure 5: (a) All clear (b) Landing direction (c) Wave off (d) Move to left. Each subfigure shows the accuracy for detecting 14 keypoints and their relationships, indicating that the suggested method can handle

a variety of human movements. An abstract yet accurate foundation for analyzing human motions is provided by the skeletal representations. For instance, the 'All clear' action is indicated by downward extended arms in subfigure (a), and the upward wave motion is seen in subfigure (c), where arms are elevated. These graphics demonstrate the variety in action recognition and demonstrate the significance of the retrieved keypoints in differentiating across actions. This skeleton architecture facilitates the efficient and successful evaluation of complex human movements.

D. FEATURE EXTRACTION

A system for recognizing actions depends significantly on the accuracy and effectiveness of the features that are extracted from the data. By capturing the significant features of the movements and poses of the human body, an efficient feature extraction procedure can significantly enhance the system's performance. A robust action recognition model was constructed in this work by extracting a variety of features. These attributes consist of MSER, angle, geodesic distance, and 3D point cloud data. The subsequent sections provide further details about each kind of feature, which is essential to truly represent human behavior.

1) KEYPOINT FEATURES: ANGLE

The angle between keypoint is crucial for understanding the orientation of limbs relative to one another during various actions. These dynamic angles vary in response to the subject's various movements, offering crucial details necessary for accurate action identification [10]. Action recognition systems may be made far more accurate by keeping an eye on these angles [39]. This work used keypoints taken out of pose estimation models to compute thirteen distinct angles. These angles track the relative locations of joints and limbs [11], enabling a thorough examination of how the body moves. Figure 5 illustrates the angles. To compute the angle between two keypoints, we utilized the formula shown in Equation (6):

$$\theta = \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) \quad (6)$$

where (x_1, y_1) and (x_2, y_2) are the coordinates of the two keypoints. This formula calculates the angle between the vector formed by these two keypoints and the horizontal axis [40]. The resulting angle provides a measure of the orientation of the limbs or joint relative to the horizontal plane. Visualization of the angle between keypoints shows in Figure 6.

2) KEYPOINT FEATURES: GEODESIC DISTANCE

In the domain of shape analysis and computer vision, geodesic distance is an essential notion, especially when working with non-rigid objects such as the human body. Rather than only calculating the straight-line (Euclidean) distance between two places [41], it refers to the shortest path between those points on a surface while capturing the geometry of the surface. Because of this property, geodesic

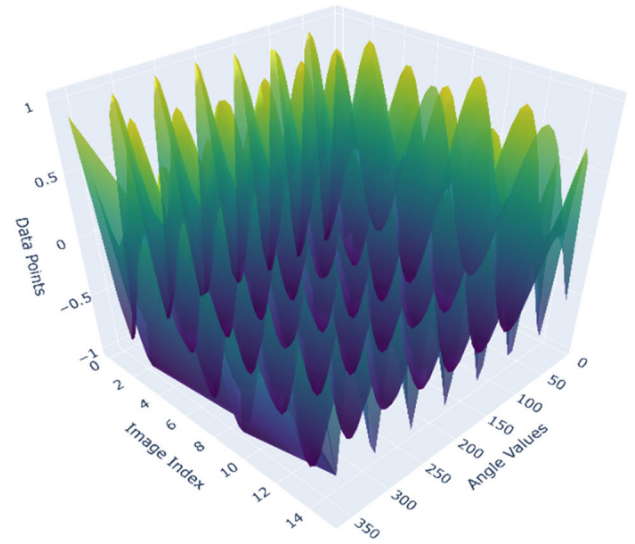


FIGURE 6. Illustration of the angle between keypoints of class all clear from UAV Gesture dataset.

distance is precious in capturing an object's internal structure, which makes it a great tool for studying human shapes in tasks involving the recognition of actions. Human action recognition depends on the ability to discriminate between various body postures and actions by examining the structure and shape of the human body. Using Euclidean distance itself frequently falls short of capturing the inherent structure of the body's position since the human body is a non-rigid shape that constantly deforms owing to movement. In contrast, the shape's inherent geometry [42], which is unaffected by bodily deformations such as bending or stretching, is captured by the geometric distance. The spatial interactions between various body parts can be precisely measured by computing the geodesic distances between keypoints on the human body. These connections can subsequently be employed as features that set various activities apart. Geodesic distance, which we use in our study because it's based on skeleton data, is the shortest path between two significant points on the human skeleton [43]. When modeling the skeleton as a graph, the vertices represent the important points (like joints), and the edges represent the connections between the vertices and points (like bones). First model the human skeleton as graph $G = (V, E)$, where V represents keypoints, and E represents edges. For each pair adjacent keypoints $p_u = (x_u, y_u)$ and $p_v = (x_v, y_v)$, calculate the Euclidean distance in Equation (7):

$$d_E(p_u, p_v) = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2} \quad (7)$$

The geodesic distance $d_G(p_i, p_j)$ between keypoints p_i and p_j is the shortest path on the graph. It is computed by finding the path that minimizes the sum of Euclidean distances in Equation (8), use dijkstra's algorithm for shortest path:

$$d_G(p_i, p_j) = \min_{p_i \rightarrow j} \sum_{(u,v) \in P_{i \rightarrow j}} d_E(p_u, p_v) \quad (8)$$

Visualization of the geodesic distance between keypoints shows in Figure 7.

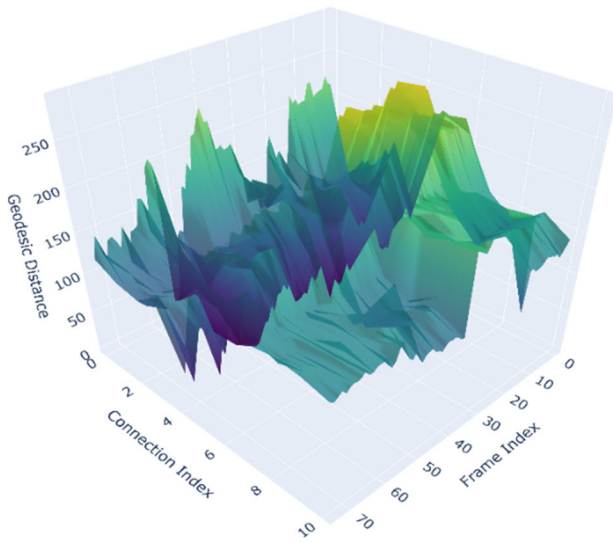


FIGURE 7. Illustration of the geodesic distance between keypoints of class all clear from UAV Gesture dataset.

3) FULL BODY FEATURES: 3D POINT CLOUD

The application of 3D point cloud features has become a revolutionary technique in the field of human action recognition, providing a detailed and extensive three-dimensional representation of the human body [44], [45]. A 3D point cloud, as compared to conventional 2D visualizations, includes the subject's scale, depth, and location in space, enabling a detailed analysis of human movement and posture. Forecasting data points over the x , y , and z coordinates produces a graphical representation of the body called a 3D point cloud [46]. Accurately identifying difficult actions needs this understanding.

In essence, a point cloud is an arrangement of distinct points in a three-dimensional coordinate system, each of which denotes a different spatial location. Each data point in this study is represented mathematically as $P_i = (x_i, y_i, z_i)$, where the coordinates in the image plane are represented by x_i and y_i and depth information is represented by z_i providing the image with the third dimension that turns the 2D silhouette into a complete 3D structure [47]. A more comprehensive understanding of body postures and gestures is made possible by converting from 2D to 3D, making action recognition more reliable and accurate.

The first step in extracting 3D point cloud feature is to locate the human silhouette in 2D images. The human figure is separated and becomes a silhouette by implementing segmentation methods like thresholding and background subtraction [48]. To ensure a clear and connected structure, morphological operations like erosion and dilation are applied, resulting in a refined silhouette.

The 3D point cloud is generated by combining the silhouette with depth information. The depth is calculated based on

pixel intensity when estimating depth using technique such as depth from shading. The relationship between pixel intensity I and depth d can be represented in Equation (9):

$$d = \alpha \cdot \frac{1}{I} \quad (9)$$

where α is a constant that adjust the scale of the depth estimation based on the intensity values. This step convert the 2D silhouette into a depth-mapped image.

The 3D point cloud is then obtained by projecting 2D silhouette points into 3D space [49]. Using the calculated depth values, silhouette points are projected into 3D space. Each pixel (x, y) is assigned a depth z , forming a 3D coordinate (x, y, z) . This projection produces the raw 3D point cloud. To manage the density and redundancy of the raw point cloud, voxel grid filtering is applied, simplifying the data by representing each voxel with a single point. This results in a simplified point cloud P' shown in Equation (10):

$$P' = \{P'_i | P_i \in P, i \in Voxel\} \quad (10)$$

Next, crucial geometrical features are extracted from the 3D point cloud. These include the surface normal, which describes the orientation of the surface at each point P_i shown in Equation (11):

$$n(P_i) = \frac{(P_i - P_j) \times (P_i - P_k)}{\|(P_i - P_j) \times (P_i - P_k)\|} \quad (11)$$

where P_j and P_k are the neighboring points. Furthermore, curvatures could give the information of the local geometry of the point cloud. The curvature K at a point can be computed as in Equation (12):

$$K(P_i) = \frac{\|n(P_i) \cdot (P_j - P_i)\|}{\|P_j - P_i\|^2} \quad (12)$$

where $n(P_i)$ is the n normal vector at P_i and P_j is a neighboring point. This curvature measure can enhance the feature set by highlighting areas of significant geometric variation. Some of the geometrical entities of interest are obtained from the extracted simplified 3D point cloud. These include: Geometry, which consists of Surface norms, which stores the orientation of the surface at the particular point, and curvature which reveals local geometrical variations helps capture detail of human figure. By integrating spatial and depth information, the 3D point cloud feature enables a detailed and dimensional understanding of human motion [50], [51], proving essential for accurate human action recognition. To better illustrate the results and highlight the features of interest, the simplified 3D point cloud is color-coded. The entire human figure is represented by different colors. For better clarity, offset values are applied to scatter the silhouette layers from the points for the separation in the plane (See Algorithm 1). Figure 8 presents the flowchart of the 3D point cloud generation, summarizing the steps from input image processing to final 3d point cloud visualization. Figure 8 shows step-by-step results, including the original 2D image and binary silhouette, depth map, raw 3D point cloud, and

the 3D point cloud result, Figure 9 shows the flowchart of 3D point cloud generation process. Figure 10 presents the final results of 3D point clouds for four different action classes;

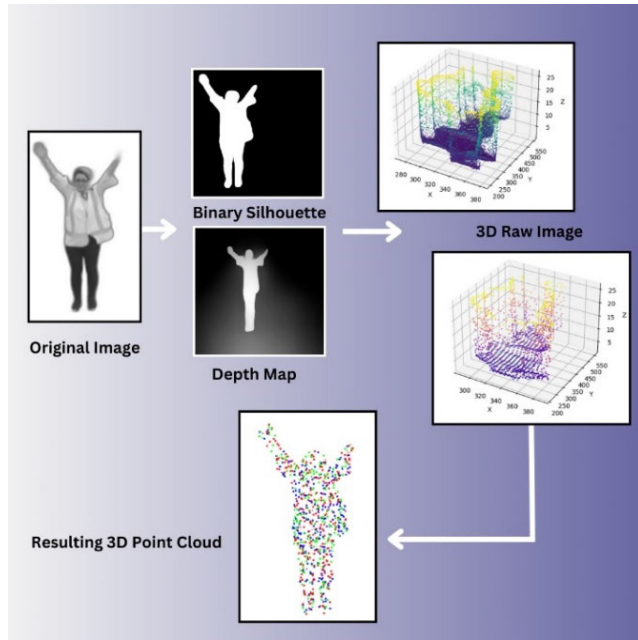


FIGURE 8. Step-by-step visualization of the 3D point cloud generation process.

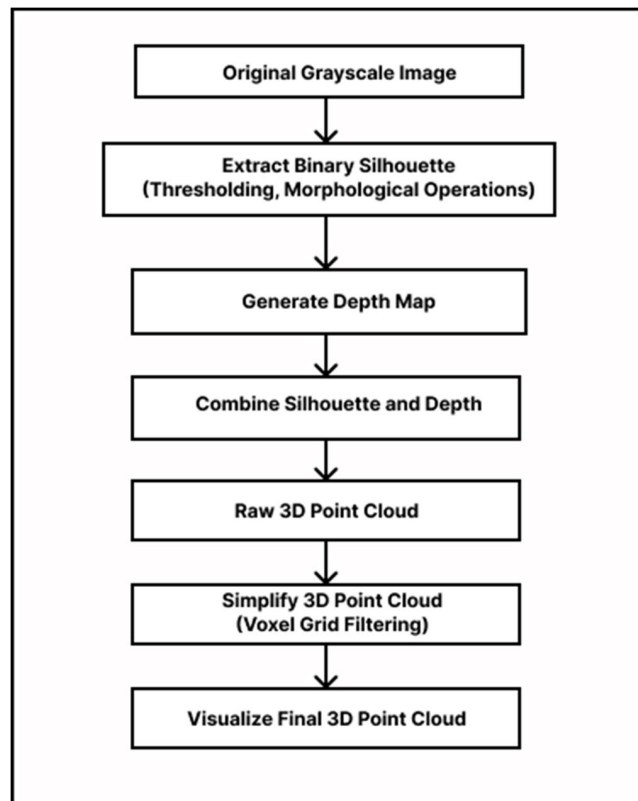


FIGURE 9. Flowchart of the 3D point cloud generation process.

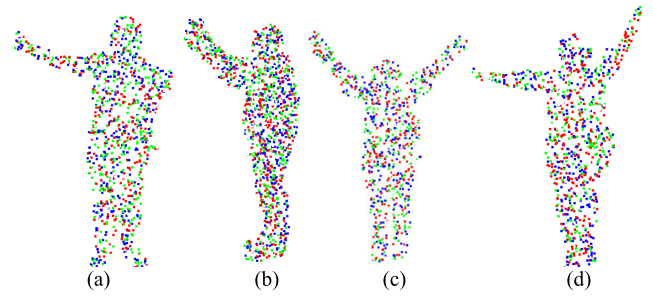


FIGURE 10. 3D point cloud of the human figure, with four different human action classes: (a) All clear, (b) Landing direction, (c) Wave off, and (d) Move to left.

4) FULL BODY FEATURE: MAXIMALLY STABLE EXTREMAL REGIONS

An approach for robust feature detection called Maximally Stable Extremal Regions (MSER) is used to identify and describe areas in a picture that remain stable in intensity over a range of thresholds. MSER is not dependent on gradient information like other feature detectors; instead, it concentrates on the stability of regions when the threshold varies. This strategy works especially well when it comes to identifying regions of consistent intensity that are important for different image-processing tasks [52], [53].

In this study, we use MSER to images in which YOLO has been used to detect human presence. Bounding boxes surrounding humans that are detected are provided by YOLO, and these boxes are used as input for MSER. We can extract stable and recognizable regions within the human figures by using MSER on these YOLO results images [54], [55]. This is important because MSER participates in finding consistent and significant elements within the recognized bounding boxes, which improves the accuracy of our action recognition system. There are multiple steps required for applying MSER: First, regions of interest inside the human figures are eliminated and concentrated using bounding boxes that Yolo recognizes [56], [57]. The MSER algorithm subsequently gives these regions to determine whether areas remain stable over a range of intensity levels. Ultimately, stable areas are taken out and examined for additional processing, which helps to differentiate between various human behaviors.

Using the stability metric $\sigma(R)$, which evaluates a region's stability as the threshold varies, the stability of an MSER region R is measure. The stability measures's formula is shown in the Equation (13):

$$\Delta(R(t)) = \frac{|R(t + \delta t) - R(t)|}{|R(t)|} \quad (13)$$

where $R(t)$ is the region at threshold t , δt is a small change in intensity, $|R(t)|$ is the area of the region. The criterion proposed is met by the maximally stable extremal regions(MSER) shown in Equation (14):

$$\Delta_{MSER}(t) = \min \Delta(R(t)) \quad (14)$$

The desired regions which satisfy the constraint should change as little as possible with the threshold t and are given by $\min \Delta(R(t))$. MSERs are also invariant to affine transformations. If an affine transformation A is operation is performed on the original image I , the identified MSER in the processed image, given in Equation (15):

$$R'(t) = A \cdot R(t) \quad (15)$$

where, $R'(t)$ represents the transformed region, preserving the properties of the original MSER under affine changes. The results of MSER feature are visualized in Figure 11.

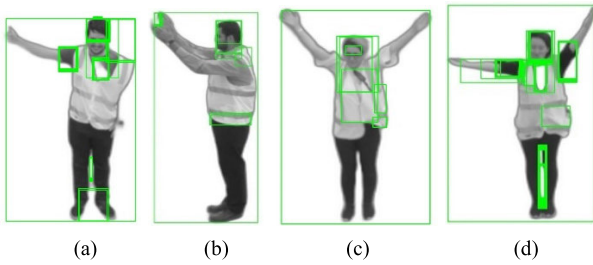


FIGURE 11. MSER feature extraction on human figure, with four different human action classes: (a) All clear, (b) Landing direction, (c) Wave off, and (d) Move to left.

Algorithm 1 3d Point Cloud

Input:

2d human image I
Human Silhouette S

Output:

3d point cloud P (A collection of points in 3d space)

Method:

Preprocess Images:

Convert the 2d human image I to a binary human silhouette S using thresholding technique
Apply morphological operations to S to enhance the silhouette

Initialize Point Cloud:

Create an empty list P to store the 3d points

Construct 3d points:

For each pixel coordinate (x, y) in the dimensions of the silhouette:

If $S[y, x] = 1$:

 Compute the depth d using the formula:

$$d = \alpha \cdot 1/I$$

 Create a point (x, y, z) and add it to the point cloud P

Post-process Point Cloud:

Apply Voxel grid filtering to reduce number of point in P

Visualize Point Cloud:

Generate a visualization of the point cloud P in 3d space

Return 3d point cloud P representing in the human in the three-dimensional space

E. QUADRATIC DISCRIMINANT ANALYSIS

An effective technique for classification tasks, quadratic discriminant analysis (QDA) can also be adapted for optimization. The most significant difference between QDA and Linear Discriminant Analysis (LDA) is the assumption regarding the covariance structure [58], [59]. Since QDA

encourages each class to have its covariance matrix, it is more versatile and able to handle complex, non-linear data distributions, which assumes that all classes share the same covariance matrix [60].

QDA is useful mainly when features show different levels of variability between classes and linear bounds are not appropriate. QDA is well-suited for tasks where capturing non-linearity is required for class distinction, like in human action recognition systems where posture and body motion vary drastically between action classes. This is when QDA forms quadratic decision boundaries [61], [62]. By improving the discriminative power between classes using quadratic decision surfaces, QDA optimizes the decision boundaries. QDA enhances the system's efficiency by capturing the complexity and non-linear correlations between various features and providing each class its covariance matrix [63], [64].

Our system leverages angles, geodesic distances, and other features as inputs, and uses Quadratic Discriminant Analysis (QDA) as an optimization tool to improve the classification accuracy of human motions. When optimizing systems [65], QDA is especially well-suited when various feature sets exhibit differing degrees of variance among action classes.

In order for the QDA optimizer to maximize the likelihood of the data given the model, it must identify the ideal parameters θ . The likelihood function $L(\theta)$ is defined as follows for a multivariate normal distribution shown in Equation (16):

$$L(\theta) = \prod_{i=1}^n P(x_i | y_i) \quad (16)$$

where x_i is the feature vector for the i -th observation, y_i is the corresponding class label, $P(x_i | y_i)$ is the class-conditional probability modeled using gaussian distributions. For each class k , the probability density function of a multivariate gaussian distribution is given by Equation (17):

$$p(x | y = k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \times \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right) \quad (17)$$

where μ_k is the mean vector of the class k , Σ_k is the covariance matrix of class k , $|\Sigma_k|$ is the determinant of the covariance matrix, d is the dimensionality of the feature vector x .

By optimizing these parameters, QDA finds the decision boundaries that separate the action classes most effectively. In our system, QDA is utilized as an optimizer to fine-tune the decision boundaries between different human action classes [66], [67]. By leveraging the unique covariance structure of each class, QDA optimizes the classification of actions based on key body motion features. The features extracted from the skeleton model and body movement are used to train the QDA optimizer which adjusts the decision surfaces to reduce misclassification and enhance overall system accuracy. Figure 12 visualizes the QDA plot.

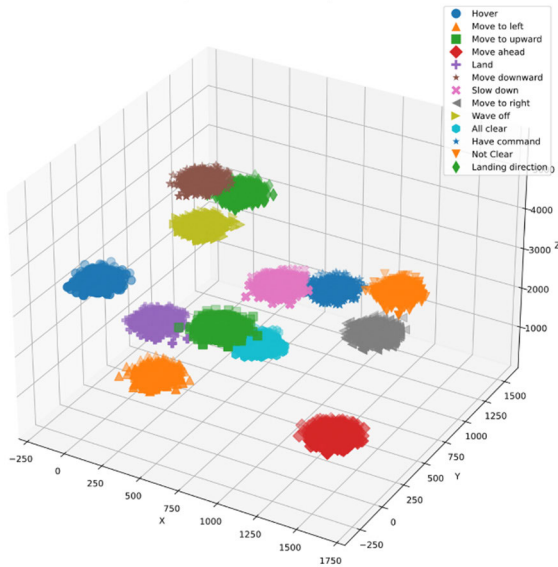


FIGURE 12. Visualization of QDA optimizer for UAV Gesture dataset.

F. DEEP NEURAL NETWORK

We employed a deep neural network (DNN) to classify human actions in drone images. Considering the capability to learn intricate data through multiple levels of interconnected neurons, DNNs can be extremely successful for complex classification problems [68], [69], [70], [71], [72], [73]. To accurately recognize actions, the model must recognize and understand high-level patterns in the data. DNN comprises several layers, each of which employs learned weights and biases to change the input data. Due to their capability to collect high-level features inside their deeper layers, DNNs are an excellent choice for tasks involving complex patterns in imagery [50], [51]. Mathematically, each layer of a DNN can be described in Equation (18):

$$h^{(l)} = f(W^{(l)}h^{(l-1)} + b^{(l)}) \quad (18)$$

where $h^{(l)}$ is the output of the l -th layer, $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector for the l -th layer, $f(\cdot)$ is the activation function (commonly ReLU: $f(x) = \max(0, x)$). For classification, the final output layer uses the softmax function to compute class probabilities Equation (19):

$$P(y = k|x) = \frac{e^{W_k^T(L)}}{\sum_{j=1}^K e^{W_j^T h(L)}} \quad (19)$$

where K is the number of classes, x is the input, and $h^{(L)}$ is the output of the last hidden layer. The network is trained using the categorical cross-entropy loss function Equation (20):

$$L = \sum_{k=1}^K y_k \log P(y = k|x) \quad (20)$$

where y_k is the true label. The DNN was optimized using backpropagation and the Adam optimizer allowing it to learn from the dataset and improve classification accuracy efficiently [30], [73], [74], [75], [76]. Adam uses the first and

second moments of the gradients to stabilize the learning process. The weight update at each iteration t is given in Equation (21):

$$W^{(l)} \leftarrow W^{(l)} - \eta \frac{m_t}{\sqrt{v_t} + \epsilon} \quad (21)$$

where $W^{(l)}$ is the weight matrix of the l -th layer, η is the learning rate, and m_t is the first moment of the gradient. Figure 13 shows the architecture of DNN.

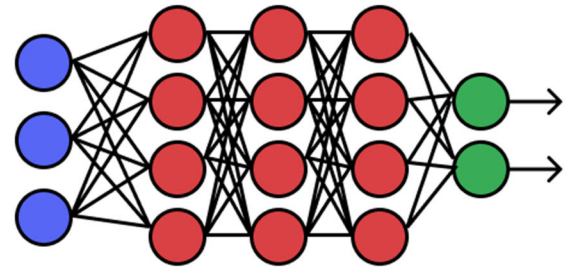


FIGURE 13. Depicting the architecture of DNN.

TABLE 1. Equipment and specifications used in experiment in the experiment.

Equipment	Model/Specifications			
Processor	Intel(R) Core(TM) i5-3210M@ 2.50GHz			
Memory	8 GB RAM			
Operating System	Windows 10, 64-bit			
Development Environment	Pycharm			
Camera	Dataset			
	UAV	UAV	UCF	
	Gesture	Human	ARG	
	GoPro Hero 4			
Camera	Black camera with an anti-fish eye replacement lens (5.4mm, 10MP, IR CUT)	Azure Kinect DK	Sanyo Xacti FH1A camera	
Drone	3DR Solo (rotorcraft UAV)	DJI Matrice 100	A 13' Kingfisher Aerostat helium balloon, used as the aerial platform for the camera.	

IV. EXPERIMENTAL SETUP AND ANALYSIS

The experiments were conducted on a laptop with the specifications mentioned in Table 1. The table also shows the key parameters and equipment used in the experiment. To assess our classifier's efficacy, we performed a thorough evaluation on three distinct datasets: UAV Human, UCF-ARG, and Drone Action. This evaluation allowed us to assess the robustness and scalability of the classifiers, given the different issues and situations that each dataset presented. We used

TABLE 2. Confusion matrix for human action recognition on the UAV Gesture dataset.

Classes	AC	HC	H	L	LD	MA	MD	ML	MR	MU	NC	SD	WO
AC	90	1	0	0	3	0	2	0	1	1	2	0	0
HC	1	91	0	2	0	1	0	2	0	3	0	0	0
H	0	0	89	4	0	1	2	1	0	0	0	0	1
L	1	0	0	92	0	0	1	2	0	2	2	0	0
LD	2	0	0	1	88	0	3	0	2	1	0	1	2
MA	1	0	1	0	0	93	0	1	2	0	1	1	0
MD	0	1	2	0	2	1	87	3	0	3	0	1	0
ML	1	0	0	0	0	1	2	87	1	0	4	2	2
MR	2	1	0	0	2	0	0	0	93	1	0	0	1
MU	1	0	0	3	0	2	0	1	0	90	2	0	1
NC	1	0	0	0	0	2	1	0	3	0	92	0	1
SD	3	0	2	1	1	0	2	0	0	0	0	91	0
WO	1	0	2	2	0	0	1	2	0	2	0	1	89
Mean accuracy = 90.15%													

*AC= All clear, HC= Have command, H= Hover, L=Land, LD= Landing direction, MA= Move ahead, MD= Move downward, ML=Move to left, MR= Move to right, MU= Move upward, NC=Not clear, SD= Slow down, WO= Wave off.

TABLE 3. Confusion matrix for human action recognition on the UAV Human dataset.

Classes	A	R	S	SD	SU	TU	W	WH
A	72	3	7	5	1	4	3	5
R	0	73	2	8	7	4	2	4
S	3	1	71	5	6	8	3	3
SD	1	5	6	74	4	5	2	3
SU	2	2	5	1	70	5	7	8
TU	4	3	2	6	3	75	3	4
W	6	3	2	5	6	4	69	5
WH	6	2	9	3	2	1	2	75
Mean accuracy = 72.37%								

* A= Applaud, R= Run, S=Salute, SD= Sit down, SU= Stand up, TU= Thumbs Up, W= Walk, WH= Wave hands.

TABLE 4. Confusion matrix for human action recognition on the UCF ARG dataset.

Classes	B	C	CL	D	J	OCT	R	T	W	WV
B	76	2	3	3	5	0	2	3	1	5
C	3	75	0	2	2	5	3	2	5	3
CL	6	1	78	4	1	1	2	1	4	2
D	2	4	1	79	2	1	1	4	0	6
J	5	1	3	2	76	0	5	3	3	2
OCT	3	5	2	3	5	75	2	2	3	0
R	1	1	2	5	2	1	77	3	5	3
T	2	2	3	0	2	0	4	78	7	2
W	2	7	4	3	2	2	3	2	74	1
WV	3	5	3	1	2	5	2	1	1	77
Mean accuracy = 76.50%										

* B= Boxing, C= Carrying, CL= Clapping, D=Digging, J= Jogging, OCT= Open close trunk, R= Running, T= Throwing, W= Walking, WV= Waving.

accuracy, recall, and precision as the performance metrics for evaluating the classifiers.

A. DATASETS DESCRIPTION

1) UAV GESTURE DATASET

The collection of video recordings developed specifically for UAV control gesture and action recognition is known as

the UAV-Gesture dataset. For UAV navigation and control in particular, this data is essential for the development of systems that understand and to human movements and actions in outdoor environments [59]. The thirteen different gestures and actions in the collection All clear, have commands, hover, land, lading direction, move ahead, move downward, move to left, move to right, move upward, not clear, slow down, and

TABLE 5. Proposed system's performance evaluation UAV Gesture dataset.

Classes	Precision	Recall	F1 score
AC	0.87	0.90	0.88
HC	0.97	0.91	0.93
H	0.93	0.91	0.91
L	0.88	0.92	0.89
LD	0.92	0.88	0.89
MA	0.92	0.93	0.92
MD	0.86	0.87	0.86
ML	0.88	0.87	0.87
MR	0.91	0.93	0.91
MU	0.87	0.90	0.88
NC	0.89	0.92	0.90
SD	0.94	0.91	0.92
WO	0.92	0.89	0.90
Mean	0.90	0.90	0.89

TABLE 6. Proposed system's performance evaluation UAV Human dataset.

Classes	Precision	Recall	F1 score
A	0.77	0.72	0.74
R	0.79	0.73	0.75
S	0.68	0.71	0.69
SD	0.69	0.74	0.71
SU	0.71	0.70	0.70
TU	0.71	0.75	0.72
W	0.76	0.69	0.72
WH	0.70	0.75	0.72
Mean	0.726	0.723	0.718

TABLE 7. Proposed system's performance evaluation UCF ARG dataset.

Classes	Precision	Recall	F1 score
B	0.74	0.76	0.75
C	0.73	0.75	0.74
CL	0.79	0.78	0.78
D	0.77	0.79	0.78
J	0.77	0.76	0.76
OCT	0.83	0.75	0.79
R	0.76	0.77	0.76
T	0.79	0.78	0.78
W	0.72	0.74	0.73
WV	0.76	0.77	0.76
Mean	0.766	0.765	0.763

wave off encompass a variety of commands and movements that can be used to operate both helicopters and airplanes. The corresponding gesture and action labels are attached to each video clip, facilitating supervised learning and evaluation. The various UAV commands represented in the dataset are crucial for training action and gesture recognition systems. The collection comprises a total of 119 video clips, amounting to 37,151 frames. Since the videos are recorded in high-definition, they ensure accurate and detailed representations of the actions and gestures.

2) UAV HUMAN DATASET

The UAV Human dataset [52] is a comprehensive benchmark developed with unmanned aerial vehicles (UAVs) to improve the development and evaluation of UAV-based models. This dataset includes 119 people in 67,428 video segments and

22,476 frames that have been specifically annotated for estimating human pose, including significant details for in-depth pose analysis. With 41,290 frames and 1,144 distinct identities as well, this dataset allows for reliable person identification in a variety of settings. It comprises 22,263 frames with annotations for attribute recognition, including accessories and clothing color. The dataset was collected over three months using UAVs operating in a variety of urban and rural settings. There is no shortage of scenarios in this collection, which covers both daytime and nighttime conditions. A wide range of subject appearances, backgrounds, lighting settings, weather fluctuations, occlusions, camera motions, and UAV flight attitudes are captured in the dataset. In this study, we utilized a subset of the UAV Human dataset that specifically focuses on 8 action classes: applaud, run, salute, sit down, stand up, thumbs up, walk and wave hands.

TABLE 8. Comparison of the proposed system with the state-of-the-art method over the UAV gesture, UAV human, and UCF ARG.

Methods	UAV Gesture (%)	UAV Human (%)	UCF ARG (%)
P-CNN[60]	91.90	-	-
MLP[76]	94.84	-	-
M-Feature CNN[10]	95.00	44.00	-
MSST- RT[74]	-	41.22	-
HDBN[79]	-	47.23	-
Optical flow[78]	-	-	68.00
SURF[63]	-	-	73.72
AP-TransNet [60]	-	-	72.70
Proposed system	90.15	72.37	76.50

3) UCF ARG DATASET

The multi-view dataset developed [16] for human action identification is called UCF-ARG (University of Central Florida-Aerial camera, Rooftop camera, and Ground camera). It comprises recordings of 12 different actors performing 10 different actions. Three cameras recorded these actions: one on the ground, one on a rooftop 100 feet above, and one installed on an aerial camera atop a 13-foot. The following actions are part of the dataset: boxing, carrying, digging, jogging, open-close trunk, running, throwing, walking and waving. The actions are captured using a high-definition camcorder (Sanyo Xacti FH₁A camera) 1920 X 1080 at 60fps.

4) EXPERIMENT 1: EXPERIMENTAL RESULTS USING PROPOSED APPROACH

On all three benchmark datasets, we have evaluated the proposed system for human action recognition (HAR). Finally, the system was evaluated on our UAV-Gesture dataset and produced a mean accuracy of 90.15%, showing its strong performance. In Table 2 we provide the confusion matrix of this dataset. The system achieved a mean accuracy of 72.37% on the UAV-Human dataset with the results in the detailed confusion matrix shown in Table 3. The more complex and varied actions of the UAV-Human dataset can be a reason for this lower accuracy. The system achieved a mean performance of 76.50% on the UCF-ARG, and the confusion matrix is shown in Table 4, proving to be competitive with several actions. In Table 5, we detail the precision, recall, and f1 score of this system in each class in the UAV-Gesture Dataset. Just as in Table 6, gives these metrics for the UAV-Human dataset, while Table 7 lists the results for the UCF-ARG dataset.

5) EXPERIMENT 2: EVALUATION USING OTHER CONVENTIONAL

We have examined the suggested human action recognition method with other existing, related systems using UAV Gesture, UAV Human, and UCF ARG datasets. Table 8 compares our method with traditional state-of-the-art procedures.

To provide a comprehensive analysis of the proposed system's performance, the ablation study is presented in Table 9.

V. DISCUSSIONS

The proposed system's performance varies on all datasets, demonstrating both its several benefits and limitations for drone-based human action recognition. The system achieves 90.15% accuracy on the UAV Gesture dataset, demonstrating good performance. The result demonstrates how well the used feature extraction methods such as 3D point cloud and MSER recognize human action from aerial images. It is challenging for the system to recognize some actions, as demonstrated by the UAV Human dataset's lower accuracy of 72.37%. Due to changing viewpoints in drone footage, action recognition can be complicated by variations in camera angles and surrounding elements. Accuracy on the UCF-ARG dataset is 76.50%, which shows an achievable performance. This dataset illustrates the ability of the system to handle an extensive number of actions by incorporating a wider range of human actions. The variation could be determined by the camera angles used to record the videos, which may impact how well the system captures important movement information. Factors such as continuously changing backgrounds, occlusions, and drone movement also contribute to the reduced performance. Working with the UAV Human and UCF-ARG datasets makes these challenges quite apparent as these complexities cause the system's performance to further drop. Despite these difficulties, the proposed system continues to perform better than a lot of cutting-edge techniques, demonstrating its overall robustness in handling drone-based human action recognition.

A. COMPUTATIONAL EFFICIENCY AND COMPARISON OF 3D POINT CLOUD GENERATION

The computational efficiency of the proposed system to generate a 3D point cloud was analyzed with the help of the hardware mentioned in the experimental setup and analysis section. The method on average processed 3 frames per second (FPS) on a regular desktop using an Intel Core i5 processor with 8GB RAM. Despite the FPS looking rather low, it shows the proposed system's effectiveness in low-resource systems conditions, highlighting that it can generate the 3D point clouds without major investments in GPUs or other computational equipment.

1) COMPARISON WITH OTHER METHODS

To contextualize the system's performance, a comparison with traditional ML/DL-based 3D reconstruction technique is provided in Table 10:

The results show the system balance between computational efficiency and performance. Unlike SFM and NeRF which can only be implemented with high-end GPUs and still consume a lot of computational resources, the proposed method performs nearly optimal on all a CPU. This makes it more appropriate to systems where hardware limitations play a major role, such as UAV-based systems.

TABLE 9. Ablation study showing the performance of the proposed system at each step.

Ablation Settings	Preprocessing	YOLO (Human Detection)	Keypoint Extraction	3D Point Cloud	MSER	Angle	Geodesic Distance	QDA Optimizer	Ectropy Classifier	UAV Gesture (%)	UAV Human (%)	UCF ARG (%)
Baseline	✓	✗	✗	✗	✗	✗	✗	✗	✓	55.14	53.26	50.01
YOLO (Human Detection)	✓	✓	✓	✓	✗	✗	✓	✓	✓	78.50	68.45	72.80
Keypoints Extraction	✓	✓	✓	✗	✗	✓	✓	✓	✓	81.50	70.10	74.00
Geodesic Distance + 3D Point Cloud	✗	✓	✓	✓	✗	✗	✓	✓	✓	85.30	71.60	75.90
3D Point Cloud + MSER	✓	✓	✓	✓	✓	✗	✗	✓	✓	86.10	72.20	76.20
Feature Fusion	✗	✓	✓	✓	✓	✓	✓	✓	✓	88.00	73.50	77.30
Full System (Without QDA)	✓	✓	✓	✓	✓	✓	✓	✗	✓	89.50	74.10	78.00
Proposed System	✓	✓	✓	✓	✓	✓	✓	✓	✓	90.15	72.37	76.50

TABLE 10. Comparison of the proposed 3D point cloud with traditional and ML/DL- based techniques.

Method	Hardware Requirements		Processing Speed (FPS)	Strengths
Proposed Method	CPU	(Intel Core i5)	3 FPS	Operates efficiently on low resources while maintaining high accuracy. Computationally intensive but accurate for multi-view setups.
Structure from Motion[83]	CPU		<1 FPS	Higher FPS but requires multiple views and computational power.
Stereo Matching[84]	CPU/GPU		1-5 FPS	High fidelity but resource-intensive.
Neural Radiance Fields (NeRF)[85]	High-end GPU		0.1-0.5 FPS	

However, for many applications, such as UAV monitoring and analysis, the achieved 3 FPS is acceptable. Furthermore, the feature points show that the system can be optimized to run at even greater frames per second (FPS) under complicated dynamic settings by either upgrading the hardware or optimizing the algorithm.

VI. CONCLUSION

In this paper, we presented an advanced human action recognition system for aerial imagery that combines YOLO

for human detection with multilevel feature fusion and a deep neural network for classification. Our approach addresses challenges unique to UAV-based action recognition, such as variable angles, complex human movements, and dynamic environmental conditions. By employing pre-processing methods and extracting significant features—including 3D point clouds, joint angles, and geodesic distances—we enhance the reliability and depth of human motion analysis in drone footage. Experimental results on the UAV Gesture, UAV Human, and UCF-ARG datasets demonstrated high accuracy rates of 90%, 62%, and 76%, respectively, underscoring the effectiveness of our method. This robust framework improves upon traditional recognition methods, enabling accurate human action identification for applications like surveillance, search and rescue, and disaster response. Future work could explore real-time processing, adaptation to more complex settings, and enhanced feature extraction to capture finer details, marking a step forward for drone-based action recognition in critical UAV applications.

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